

A PROJECT TITLE
ON
Air Pollution Anomaly Detection Using DBSCAN and Isolation Forest
project report submitted in partial fulfilment of the requirements for the Course
MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING.

Name of Title: Air Pollution Anomaly Detection Using DBSCAN and Isolation Forest

1. Abstract:

Air pollution is a critical environmental issue that significantly impacts public health, ecosystems, and urban sustainability. The continuous collection of air quality data from monitoring stations generates large and complex datasets, making the manual identification of abnormal pollution events both challenging and inefficient. This project proposes an Air Pollution Anomaly Detection System that employs unsupervised machine learning techniques to identify anomalous air quality patterns without relying on labelled data.

The system utilizes key environmental parameters, including PM2.5, PM10, CO, NO₂, SO₂, O₃, temperature, and humidity, obtained from publicly available air quality datasets. The collected data is pre-processed through data cleaning, missing value handling, feature selection, and feature scaling to improve model reliability and performance. Two anomaly detection algorithms, DBSCAN (Density-Based Spatial Clustering of Applications with Noise) and Isolation Forest, are implemented and comparatively analysed. DBSCAN detects anomalies by identifying low-density observations outside meaningful clusters, while Isolation Forest isolates anomalous instances through random tree-based partitioning.

The proposed system follows the complete machine learning lifecycle, including data preprocessing, model training, evaluation, model serialization, and deployment through a lightweight web interface for real-time predictions. By enabling the early detection of unusual pollution events, the system demonstrates the practical application of anomaly detection techniques in environmental monitoring and supports timely, data-driven decision-making for smart city and public health initiatives.

2. INTRODUCTION:

Air quality sensors continuously collect environmental data, but sudden pollution spikes can occur because of industrial emissions, vehicle congestion, fires, or sensor malfunctions. The project aims to automatically detect such abnormal events using unsupervised machine learning, without requiring labelled data. DBSCAN and IsolationForest are used to identify and compare unusual pollution patterns.

3. SOFTWARE REQUIREMENTS:

Python 3.x, Google Colab / Jupyter Notebook, Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn.

- Python 3.x – Used as the main programming language to develop the ML project.
- Google Colab / Jupyter Notebook – Used to write, run, and test the Python and ML code.

- Pandas – Used to load, clean, and preprocess the dataset.
- NumPy – Used for numerical calculations and handling arrays.
- Scikit-learn – Used to train, test, and evaluate machine learning models.
- Matplotlib – Used to create graphs and visualize data and results.
- Seaborn – Used to create statistical graphs such as correlation heatmaps.

Signature of the Guide

Course Instructor